Christopher P. Ravosa

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SUMMARY

Software engineer with experience building distributed backend systems, AI infrastructure, simulation platforms, and real-time applications across cloud and gaming technologies.

SKILLS

C# • TypeScript • Python • Java • C++ • PostgreSQL • MongoDB • Git • REST APIs • React • Docker • AWS • GCP

EXPERIENCE

Moku, ***Senior Software Engineer***
November 2025 - June 2026

- Promoted to lead technical execution of AI simulation infrastructure and gameplay systems
- Provisioned Linux VM infrastructure on Google Compute Engine for AI simulation and debugging
- Coordinated engineering efforts to implement gameplay systems for 2D and 3D Unity games

Moku, ***Software Engineer***
Mar 2025 - November 2025

- **Reduced AI model iteration cycle by 60 days** by designing a new reinforcement learning infrastructure
- Built a deterministic replay pipeline for debugging, auditing, and reproducing execution of simulation sessions
- Utilized Python, C#, JSON and YAML configuration files to create configurable agent training environments

Major League Baseball, ***Associate Software Engineer***
Aug 2023 - Mar 2025

- Led development of an interactive software product when lead engineer was reallocated to another project
- Designed, authored, and deployed backend APIs supporting two live Unity applications
- Developed and owned real-time TCP multiplayer servers supporting live user interactions
- Built pipelines for distributing assets via CDN, enabling dynamic content delivery while reducing client footprint
- **Diagnosed and resolved production outage in 15 minutes** while participating in on-call rotations
- Created server stress testing frameworks for identifying integrity issues and improving reliability under load
- **Reduced application memory usage by 40%** by developing compute shaders to generate textures at runtime
- Authored spatial computing systems to visualize real-time baseball events in 3D space on Apple Vision Pro

Naughty Dog, ***Associate Technical Designer***
Jul 2022 - Jun 2023

- Built networked systems with host-authoritative architecture, ensuring state consistency across distributed clients
- Implemented synchronization logic resilient to latency, packet loss, and unreliable network conditions
- Designed and tested systems to prevent state divergence and maintain consistency in real-time environments

Activision, ***Design Intern (Games, Systems, UX)***
Summer 2020 & Summer 2021

- Designed aerial combat vignettes and collaborated across art and animation teams to implement scenes
- Tested and debugged gameplay systems across singleplayer campaign levels

PROJECTS

Real-Time Climate Monitoring System

- Built an IoT device and fault-tolerant, climate monitoring firmware for an ESP32 microcontroller using ESP-IDF
- Implemented bidirectional WebSocket communication between firmware, cloud-hosted backend, and browsers
- Designed and built a web dashboard for end users to view data and remotely control the IoT device

Chasm Master

- Built an LLM-powered semantic validation service that maps free-form player input to predefined game actions
- Automated server deployment using GitHub Actions to support a playable demo on ChrisRavosa.com

EDUCATION

Marist College

- MSc • ***Software Development*** • Concentration in **Mobile Computing**
- BSc • ***Computer Science*** • **GPA 3.87**